PRANAV MANOJ HIRAWAT

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SUMMARY

Final-year BCA student at PES University who builds backend systems in Python. Built the decision-making control plane for an API security gateway — an agent loop that correlates attack evidence, decides policy, and integrates a local LLM for incident reporting. Seeking a backend or AI engineering internship.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (basics)

Backend & Data Stores: Redis, PostgreSQL, MySQL, REST APIs

AI & ML: LLM integration (Ollama), Scikit-learn, NLTK, Pandas, NumPy

Testing & Tools: pytest, Git, Linux, Jupyter, Tableau

PROJECTS

Intelligent API Security Gateway — Agentic Control Plane

Python · Redis · Ollama · pytest — team project; built the Python control plane

- Built the control plane of an API security gateway: an agent loop running every 30 seconds that reads security events from Redis, decides what to do about them, and writes policy the gateway enforces.
- Wrote the correlation logic that groups unrelated IP addresses into a single named attack campaign using shared traits — target endpoint, User-Agent, subnet and overlapping timing — with a confidence score per campaign.
- Implemented the policy engine mapping campaign severity and confidence to an action ladder (monitor, throttle, temporary block, escalate), written to Redis with a TTL so decisions expire without manual cleanup.
- Integrated a local LLM (Ollama) for incident summaries, with a null-provider fallback so the system runs fully without an LLM available.
- Placed the LLM strictly downstream of enforcement, so a prompt-injected summary can mislead a human reader but cannot alter a blocking decision. Covered by 236 tests, including a negative scenario that must produce zero detections.

Fake Review Detection System

Python · Scikit-learn · NLTK · TF-IDF

- Built a supervised classification pipeline using TF-IDF vectorization to identify fraudulent product reviews in e-commerce data.
- Compared multiple algorithms and tuned hyperparameters to select the final model.

Healthcare Data Analysis

Python · Tableau

- Cleaned and analysed multi-variable patient data, then built Tableau dashboards showing relationships between diabetes and cholesterol markers to help identify high-risk groups.

EDUCATION

PES University, Bangalore — Bachelor of Computer Applications (BCA) Expected May 2027 | CGPA 7.66 / 10

POSITIONS OF RESPONSIBILITY

- **Technical Core Team**, TAMS — PES University
- **Logistics Team**, Chords (inter-college music festival) — coordinated event logistics across vendors and performers